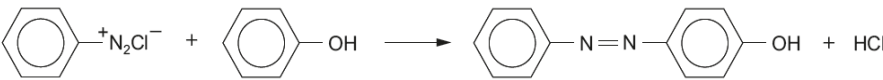


Question				Marking details	Marks available																
					AO1	AO2	AO3	Total	Maths	Prac											
10.	(a)			<table><tr><th>Stage</th><th>Reagent used</th><th>Structural formula of product</th></tr><tr><td>1</td><td>KBr / H₂SO₄ or SOCl₂ / PCl₅</td><td>CH₃CH₂CH₂Br CH₃CH₂CH₂Cl</td></tr><tr><td>2</td><td>KCN</td><td>CH₃CH₂CH₂CN</td></tr><tr><td>3</td><td>LiAlH₄</td><td></td></tr></table>	Stage	Reagent used	Structural formula of product	1	KBr / H ₂ SO ₄ or SOCl ₂ / PCl ₅	CH ₃ CH ₂ CH ₂ Br CH ₃ CH ₂ CH ₂ Cl	2	KCN	CH ₃ CH ₂ CH ₂ CN	3	LiAlH ₄		1	2	3		
				Stage	Reagent used	Structural formula of product															
				1	KBr / H ₂ SO ₄ or SOCl ₂ / PCl ₅	CH ₃ CH ₂ CH ₂ Br CH ₃ CH ₂ CH ₂ Cl															
				2	KCN	CH ₃ CH ₂ CH ₂ CN															
3	LiAlH ₄																				
	(b)	(i)		sodium nitrate(III) / nitrite and acid e.g. HCl	1			1		1											
		(ii)		 accept other correct alternative equations		1		1													
	(c)			the statement is true as HCl will be removed as it is formed, moving the position of equilibrium to the right accept other valid reasons			1	1													

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
	(d)			7.87 g of the amino acid gives 1470 cm ³ of N ₂ gas (1) n(N ₂) = 1470 / 24.5 × 1000 = 0.060 therefore 0.060 mol of the amino acid molar mass of the amino acid = 7.87 / 0.060 = 131 (1) R—CH(NH ₂)COOH → 131 and CH(NH ₂)COOH → 74 therefore R → 131 – 74 = 57 (1) it is a straight chain compound so R is CH ₃ CH ₂ CH ₂ CH ₂ therefore amino acid must be CH ₃ (CH ₂) ₃ CH(NH ₂)COOH (1)		2			2													
	(e)	(i)		2-methylpropane-1,3-dioic acid accept 2-methylpropanedioic acid	1			1														
		(ii)		<table border="1"><thead><tr><th>Protons</th><th>Splitting pattern</th><th>Relative peak area</th></tr></thead><tbody><tr><td>–CH₂–CH₂–</td><td>singlet / no splitting</td><td>4</td></tr><tr><td>–C–H–</td><td>quartet</td><td>1</td></tr><tr><td>–CH₃</td><td>doublet</td><td>3</td></tr></tbody></table> award (1) for each correct row	Protons	Splitting pattern	Relative peak area	–CH ₂ –CH ₂ –	singlet / no splitting	4	–C–H–	quartet	1	–CH ₃	doublet	3		3		3		
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				Question 10 total	3	8	3	14	2	1												